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Cretaceous and Tertiary paleomagnetic results from Southeast China and their tectonic implications

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ABSTRACT

We report Cretaceous and Tertiary paleomagnetic data from Fujian, Guangxi and Guangdong provinces in south China. In Guangxi Province we sampled Early Cretaceous redbeds of the Xinlong Formation at eight localities separated by up to 100 km. The high-temperature component obtained by thermal demagnetization is of dual polarity and passes the fold test, and implies 18° ± 8° counterclockwise rotation and insignificant poleward displacement with respect to cratonic localities in Sichuan Province. A Tertiary basin lying in a southeast-trending fault zone is also rotated counterclockwise by 40° ± 8°. Both results can be explained very well by a simple block rotation model for the Cenozoic tectonic evolution of western Guangxi Province that is driven by left-lateral shear on southeast-trending faults. The timing and sense of shear are the same as for India–Asia collision models involving the extrusion of Indochina along the Red River fault zone, which lies several hundred kilometers to the southwest, but the magnitude of fault displacement in Guangxi is much less. In the coastal provinces of Fujian and Guangdong we find Late Cretaceous directions that are rotated clockwise with respect to Sichuan Province. Combined with similar clockwise rotations found by two other studies of coeval rocks from the same region, the mean rotation for the coastal provinces is 12° ± 8°; this could indicate a coherent block rotation of the region driven by extrusion tectonics. However, the limited number of sampling localities also allows the possibility that the directions reflect local rotations due to dextral shearing driven by circum-Pacific tectonics.

1. Introduction

China is composed of tectonic blocks that have accreted throughout the Phanerozoic [e.g., 1]. The eastern half of the country is dissected by east–west trending sutures within the Qilian and Qinling foldbelts that demarcate the Sino-Korean craton from the South China Block (SCB) (Fig. 1). These landmasses appear to have begun colliding at the end of the Paleozoic near the eastern end of the suture and then scissored together [2,3].

The tectonic history of the SCB is poorly understood. Opinions differ as to whether it has been a single unit since the Precambrian [4,5] or Early Paleozoic [1,6,7], or whether it is composed of a stable nucleus in the west (the Yangtze craton) that grew progressively eastwards by accretion throughout the early and middle Mesozoic [8–12]. Despite the uncertainties in the accretionary history of the SCB, paleomagnetic evidence has established that its amalgamation was complete by the Cretaceous [e.g., 13]. A characteristic feature of the paleomagnetic poles from the SCB and other blocks of China is that they are streaked about a small circle, suggesting that local or regional rotations have occurred since the Cretaceous [13]. Many of these rotations may have been caused by the collision between India and Asia, an event which appears to have pro-

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duced significant deformation over thousands of kilometers in east Asia [14].

Seismicity and structural trends attributed to the collision have been interpreted in terms of a polyphase extrusion model, in which large tectonic blocks are squeezed eastwards away from the deformation front [15,16,41]. To the southeast of the front, Indochina is thought to have escaped the collision by moving to the southeast along the Red River fault zone. According to the extrusion model, from 50 to 17 Ma, 800–1000 km of sinistral motion occurred in the fault zone, Indochina rotated 20–25° clockwise, and the South China Sea opened. After 17 Ma, motion along the fault changed from sinistral to dextral. This second phase was accompanied by the end of seafloor spreading in the South China Sea and the initiation of strike-slip faulting farther north in China.

However, mechanisms other than the India–Asia collision may be responsible for crustal deformation in China. In eastern China, Xu et al. [17] point to evidence for north- to northeast-trending left-lateral faults, active from the Triassic to the Early Cretaceous, that were reactivated in the opposite sense in the Late Cretaceous concurrently with the formation of extensional basins. They argue that the change in the stress regime arose from a change in relative motion between the continent and the Pacific Ocean plate. Taylor and Hayes [18] interpret the opening of Late Cretaceous, north–south trending extensional basins located off the continental margin in the South China Basin to be due to the relaxation of a previously active Andean-type subduction zone. A later switch in the trend of extensional basins from north–south to east–west in the Eocene they attribute to global plate reorganization, including both changes in Pacific plate motion and the collision of India with Asia.

Here we report Cretaceous and Tertiary paleomagnetic results obtained over a large area in south China from Guangxi, Fujian and Guangdong provinces (Fig. 1). We then compare the results to an average Cretaceous pole from Sichuan Province. Sichuan is only weakly deformed, so comparison of the results reported here with those from Sichuan provide insight into the post-Cretaceous deformation of south China. By collecting samples both near the Red River fault and near the continental margin we hoped to clarify whether the India–Asia collision and/or the circum-Pacific tectonic regime (or neither) were responsible for the deformation.

2. Geology and sampling

Figure 1 shows four areas where we obtained paleomagnetic cores for this study, and Table 1 lists the bedding attitudes and locations of the sampled strata discussed below. The samples from Guangxi Province (Areas 1 and 2, Fi. 1), lie only a few hundred kilometers east of the Red River fault. Those from Area 1 lie in a large basin that trends perpendicular to the fault, and those from Area 2 lie in a narrow basin that trends parallel to the fault (Fig. 2). Late Cretaceous rocks sampled in Fujian and Guangdong provinces (Areas 3 and 4, Fig. 1) come from two of the many small basins that are oriented approximately parallel to the coast (Fig. 3).

We drilled 72 paleomagnetic cores at eight localities in the Early Cretaceous Xinlong Formation in two large basins south and east of Nanning (the capital of Guangxi Province) (Fig. 2). Samples from localities L19, L20, L102 and L103 were collected along a newly constructed road that connects Nanning and Qinzhou, and those from localities L130, L131 and L132 were collected along a road that connects Nanning and Lingshan. Sampling locality L138 lies on the road that connects Yulin and Guixian.

The Xinlong Formation varies from 80 to 2445 m in thickness and consists mostly of red or purplish red, fine- to coarse-grained sandstones and siltstones. Fossils identified in the redbeds include pelycepods (*Trigonioides kodarai*, *Nokamuranaia chingshanensis* and *Nippononaia subrotunda*) and vertebrates (*Asiatosaurus kwangshiensis*, *Prodeinodon kwangshiensis* and *Sinopliosaurus fusuiensis*), as well as plants, ostracods, estherias, gastropods and charophytes, indicative of Early Cretaceous time [19].

We also drilled 17 cores at two localities (L106 and L107) from Late Cretaceous redbeds known as the Upper Series deposits. The localities are 500 m apart and lie north of Qinzhou city (Fig. 2). The strata are red, coarse-grained sandstones and conglomerates that unconformably overlie Paleozoic sediments. No fossils have been found in the strata and the age is based on correlation with

other beds with similar lithologic characteristics [19].

In Area 2 (Fig. 1), 150 km to the northwest of Area 1, we collected 17 cores of lower Tertiary fluviolacustrine sediments from two localities (L113 and L119) in western Guangxi Province. The samples were collected along the road within the Youjiang Basin that connects Nanning with the cities of Tiandong and Tianyang. (Fig. 2 and Table 1). The elongate basin is 3–4 km deep, lies within the Youjiang fault zone, and is an important hydrocarbon producer.

Locality L113, located on the eastern flank of the basin, consists of Paleocene to lower Eocene purple-red conglomerates and conglomeratic sandstones of the Liuchou Formation. The Liuchou Formation is 34–367 m thick and contains ostracods such as *Ilyocyorus errabundis* and *Lymnocythera* spp. [19]. Samples from locality L119 come from the upper Eocene Nadu Formation,

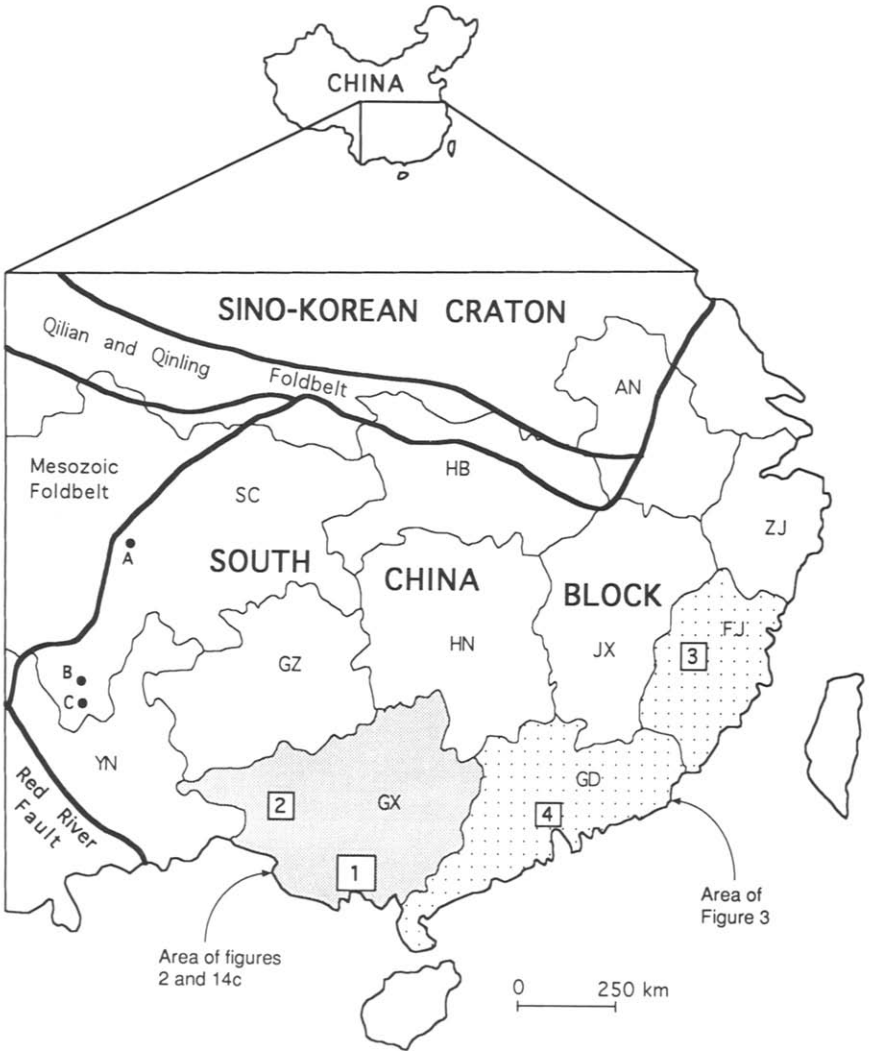


Fig. 1. Regional geologic map of south China. Tectonic boundaries from Zhang et al. [1]. Numbers in boxes are areas where paleomagnetic samples were obtained for this study. A, B, and C represent the localities sampled by Enkin et al. [34], Zhu et al. [33] and Huang and Opdyke [35] respectively (Table 2). Provinces are abbreviated as follows: *AN* = Anhui; *FJ* = Fujian; *GD* = Guangdong; *GX* = Guangxi; *GZ* = Guizhou; *HB* = Hubei; *HN* = Hunan; *JX* = Jiangxi; *SC* = Sichuan; *YN* = Yunnan; *ZJ* = Zhejiang.

TABLE 1: Summary of paleomagnetic data

Study/locality	Age	L_{lat}/L_{long}	S/D	Coord.	n/N	Pol.	D(°)	I(°)	α_{95}(°)	k
Xinlong Formation										
L19	K_1	22.5/108.5	227.7/10.0	G	10/10S	N	1.1	43.8	7.1	47.7
				S	10/10S	N	355.6	36.2	7.1	47.7
L20	K_1	22.4/108.6	285.0/16.5	G	11/11S	N	348.9	52.4	4.9	87.4
				S	11/11S	N	355.3	37.2	4.9	87.4
L102	K_1	22.3/108.3	295.0/24.0	G	10/10S	N	347.3	59.8	9.6	26.5
				S	10/10S	N	359.6	37.5	9.1	29.4
L103	K_1	22.3/108.4	280.0/40.0	G	10/10S	N	315.8	63.8	7.0	48.8
				S	10/10S	N	343.7	30.4	6.7	53.4
L130	K_1	22.7/108.6	105.0/20.0	G	8/8S	N	357.1	36.0	11.3	25.0
				S	8/8S	N	348.9	54.5	11.3	25.0
L131	K_1	22.7/108.7	265.0/35.0	G	8/8S	N	351.8	69.1	14.8	15.0
				S	8/8S	N	355.2	36.9	14.1	16.4
L132	K_1	22.7/109.3	11.0/18.7	G	6/6S	M	342.9	29.5	15.7	19.1
				S	6/6S	M	354.3	37.3	15.7	19.1
L138	K_1	22.9/109.8	264.0/10.0	G	9/9S	N	18.0	49.1	5.3	96.3
				S	9/9S	N	14.8	40.7	5.5	88.3
Mean HTC				G	8/8L	M	352.0	51.5	11.9	22.6
				S	8/8L	M	355.9	39.1	6.8	67.9
Mean HTC (-L138)				G	7/8L	M	348.1	51.3	12.9	22.9
				S	7/8L	M	353.3	38.7	6.4	90.4
Mean HTC (-L138, +L107)				G	8/9L	M	347.2	48.3	12.5	20.6
				S	8/9L	M	352.8	38.2	5.5	101.1
Upper Series (L107)	K_2	22.2/108.7	40.0/14.0	G	9/9S	N	343.3	27.2	9.3	31.6
				S	9/9S	N	349.3	35.3	9.3	31.7
Youjiang Basin										
L113	E_1	23.5/107.3	160.0/77.7	G	9/9S	R	183.7	-24.5	4.7	122.3
				S	9/9S	R	140.2	-24.9	4.7	122.8
L119	E_1	23.7/106.8	307.0/49.4	G	8/8S	R	102.7	-43.1	9.1	38.3
				S	8/8S	R	152.9	-42.3	9.1	38.3
Mean HTC				G	17/17S	R	331.7	40.4	17.2	5.3
				S	17/17S	R	325.5	33.2	6.5	31.5
Shaxian Formation										
L151	K_2	26.0/117.5	160.0/25.0	G	9/9S	N	42.0	28.1	8.8	35.2
				S	9/9S	N	23.5	38.8	7.8	45.0
L159	K_2	26.0/117.3	350.0/35.0	G	11/11S	N	352.5	51.6	7.7	36.1
				S	11/11S	N	30.2	41.3	7.4	38.8
Mean HTC				G	20/20S	N	19.2	43.8	10.8	10.1
				S	20/20S	N	27.1	40.2	5.1	41.4
Sanshui Formation (L98)	K_2	23.1/113.3	328.1/29.3	G	19/19S	N	25.5	50.7	5.1	44.1
				S	19/19S	N	36.6	27.1	4.6	53.6

K = Cretaceous, E = Cenozoic, 1 = early, 2 = late. L_{lat} and L_{long} = locality latitude and longitude. S/D = average strike and dip of the locality (note that strikes and dips may vary within a locality; the dip direction is the strike plus 90°). *Coord.* = coordinate: G = *in-situ* (geographic) coordinates; S = tilt-corrected (stratigraphic) coordinates. n/N = number of samples (S) or localities (L) demagnetized/number of samples or localities used to calculate mean direction. *Pol.* = polarity: N = normal; M = mixed; R = reversed. D = declination; I = inclination; α_{95} = 95% confidence radius for mean direction; k = best estimate of the precision parameter

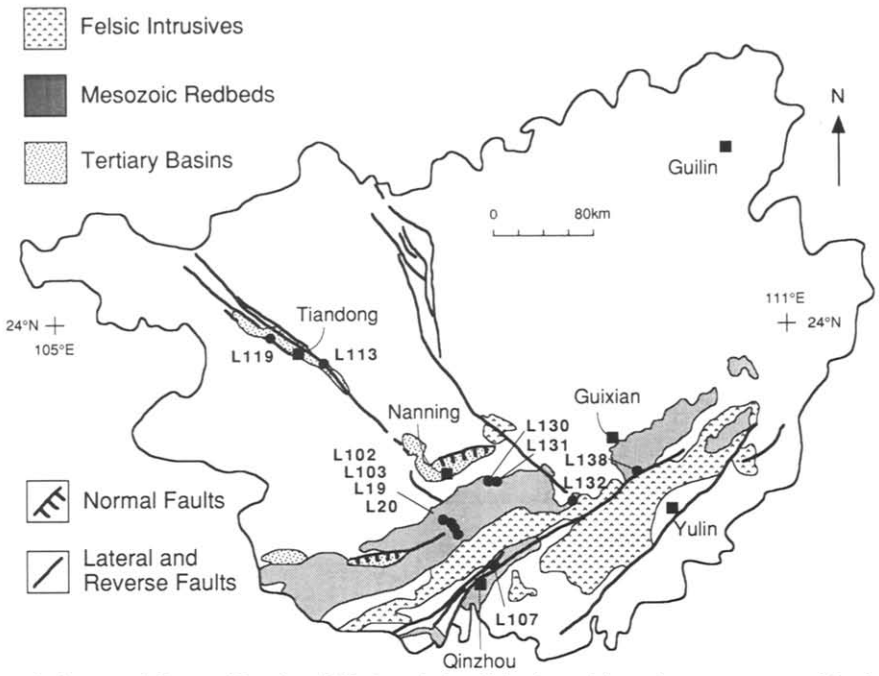


Fig. 2. Simplified geologic map of Guangxi Province [19]. Sample localities from this study are represented by dots. Zhao et al. [32] sampled Eocene sediments from the basin in and around Nanning.

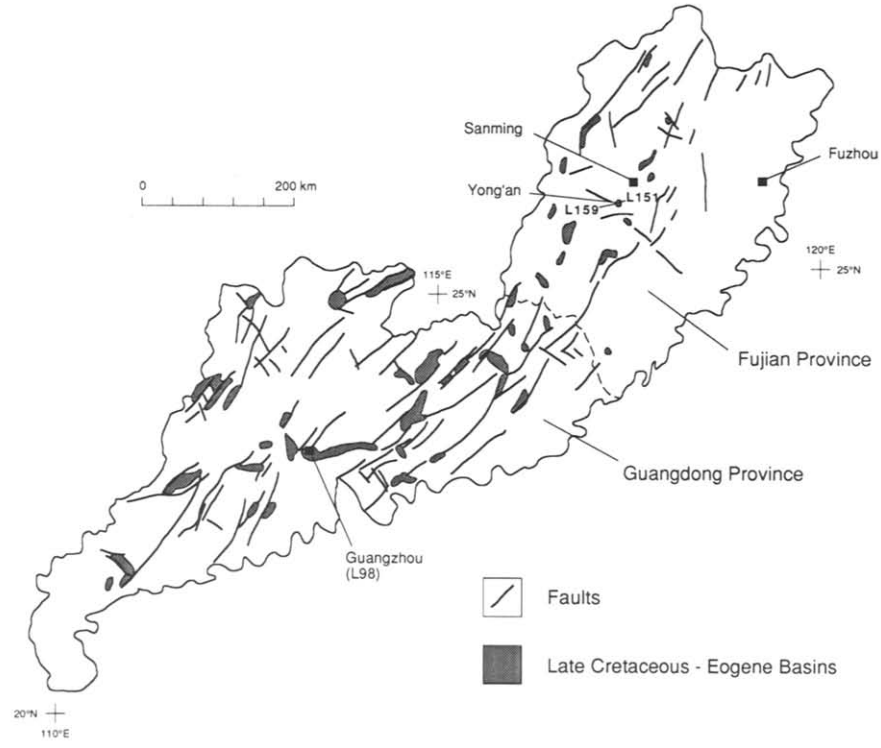


Fig. 3. Simplified geologic map of Fujian and Guangdong provinces showing distribution of Late Cretaceous to Eogene basins [45]. L98, L151 and L159 are the sample localities from this study. Zhai et al. [28] sampled the Late Cretaceous Hekou Formation from the basin in and around Yong'an.

which consists of light gray and gray-green mudstones, calcareous mudstones and siltstones that range in thickness from 167 to 679 m. Fossils found in the strata include the gastropod *Viviparus margartaeformis* and the vertebrate *Guilestes acares*. This locality lies along the south central part of the basin where the Eocene strata are in fault contact with Triassic limestone.

In Area 3 (Fig. 1), we drilled two localities (L151 and L159) of upper Cretaceous redbeds from the Shaxian Formation. The redbeds were deposited in a small basin located in and around Yong'an City in west central Fujian Province (Fig. 3). Outcrops were drilled to the east and west of Yong'an where the main road passes through the city. The upper Cretaceous rocks consist of red, coarse- to fine-grained sandstones and conglomerates deposited in a fluviolacustrine environment. Plant fossils (*Frenelopsis* and *Pseudofrenelopsis*), ostracods (*Tongxiella*) and charophytes (*Obtusochara* and *Aclistochara*) found in the Shaxian Formation indicate a Late Cretaceous age [20].

We also sampled one locality (L98) from the upper Cretaceous Sanshui Formation near Shiliugang Hospital (Chigangdong District) in Guangzhou City (Canton), Guangdong Province (Fig. 3). Like most Late Cretaceous deposits in south China, the Sanshui Formation is made up of red, fluviolacustrine coarse clastics. The presence of fossil eggs (*Oolithes elongatus*, *Paraspheroolithus* and *Elongatoolithus*), ostracods (*Cristocypridea fabarua*, *Cr. amoena*, *Cypridea cavernosa*, and *Candona declivis* and charophytes (*Charites guan-*

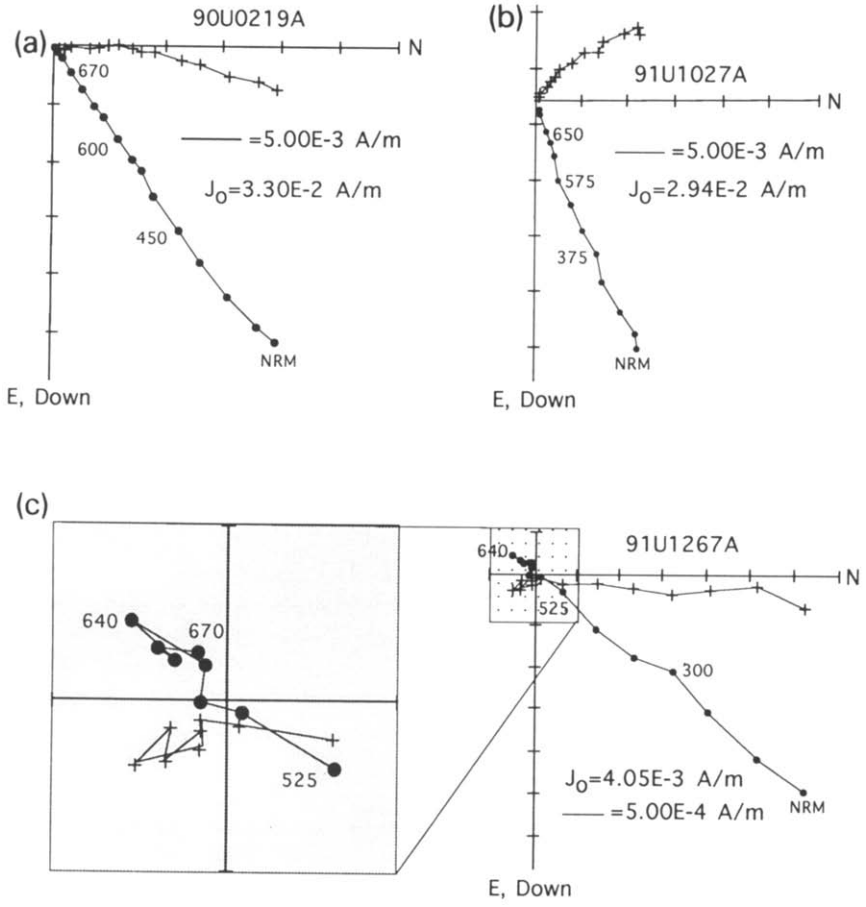


Fig. 4. Thermal vector demagnetization diagrams of samples from the Xinlong Formation. Figures (a), (b) and (c) are from localities L19, L103 and L132 respectively (*in-situ* coordinates). ● = vertical component; + = horizontal component; temperature steps are in degrees Celsius.

pingensis, *Obtusochara culmina*, *Sphaerochara granulifera* and *Mesochara xinanensis*) indicate that the Sanshui Formation is Late Cretaceous [21].

We collected our samples using a portable gasoline-powered drill, oriented them with a magnetic compass, and took sun compass bearings to correct for any local declination anomaly. In each area we took care to leave enough stratigraphic separation between cores to average out secular variation effects. We thermally demagnetized the samples in an oven built at the University of California at Santa Cruz (UCSC) that is housed in a shielded room. Calibration experiments show that the temperature of the samples in the oven can be controlled within ± 1°C and are accurate within ± 8°C of the set temperature, and that the ambient field in the cooling chamber is less than 10 nT. We demagnetized the samples in eight to twenty-one temperature steps and measured them on a 2G cryogenic magnetometer, using principal component analysis [22] and Fisher [23] statistics to analyze the results.

3. Paleomagnetic results

3.1 Early cretaceous xinlong formation

Figure 4 shows the response to thermal demagnetization for three samples representative of the 72 samples studied. The demagnetization diagrams in Fig. 4a and b exhibit two components of magnetization, a low-temperature component (LTC) (generally below 567°C) that does not trend toward the origin, and a high-temperature component (HTC) (567–680°C) that does. Most of the samples have normal polarity, except for three of the six samples from locality (L132) that are reversed. During demagnetization of these samples, a north and down component is removed and the directions nearly pass through the origin as they trend to the reversed state (Fig. 4c). The high unblocking temperature of the remanence and its resistance to removal by an alternating field (af) show that it is carried by hematite.

Table 1 lists the results of the paleomagnetic study for the Xinlong Formation. The best estimate of the precision parameter (k) for the HTC is three times greater at 100% unfolding (k = 67.9) than at 0% unfolding (22.6). The increase in clustering of the HTC, which is graphically portrayed in Fig. 5, is significant at the 95% confidence level [24] and demonstrates that the HTC was acquired before folding. From our field investigations the Late Cretaceous strata in the area are more gently folded than the Early Cretaceous rocks; thus we interpret the HTC to be a near-primary magnetization.

Note that locality L138 is the only one without a northwest-trending direction (Table 1). It will be arged later that this has tectonic significance and that this locality should be omitted when

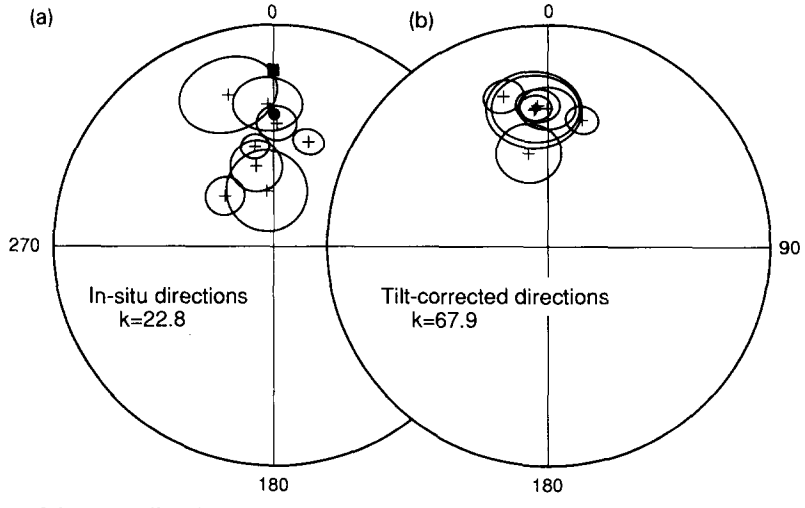


Fig. 5. Equal-area plots of the mean directions and their 95% circle of confidence from the Xinlong Formation before (a) and after (b) unfolding. ■ in (a) represents the present field direction; ● = geocentric axial dipole direction for the present latitude.

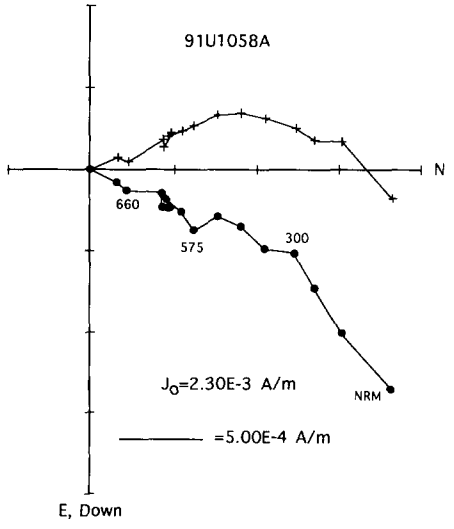


Fig. 6. Thermal vector demagnetization diagram for sample 91U1058A from the Upper Series (L107) (*in-situ* coordinates). ● = vertical component; + = horizontal component; temperature steps are in degrees Celsius.

calculating the mean HTC direction. Furthermore, if the results from the Late Cretaceous Upper Series deposits (see below) are included, the regional fold test would be significant at greater than the 99% confidence level [24] (Table 1).

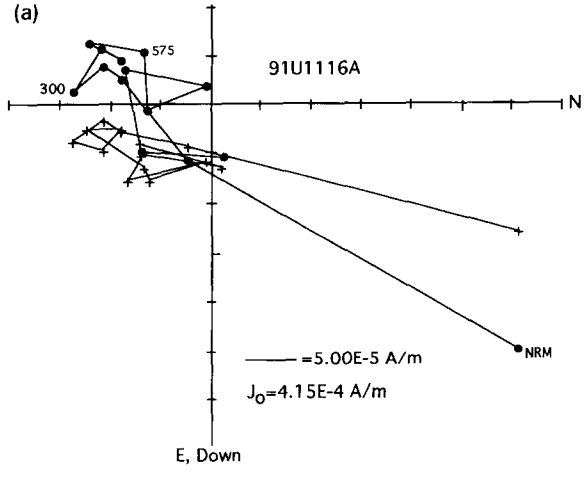
Interpretation of the low-temperature component (LTC) is not so straightforward. The maximum k value (53) occurs around 50% unfolding, but a comparison of the direction at 50% unfolding ($D = 0.5°$, $I = 49.5°$) with the expected Cretaceous direction ($D = 14.5°$, $I = 34.5°$) shows that the two directions are distinct at 95% confidence [25]. Thus, it is unlikely that this component was acquired during Cretaceous folding, and its origin remains problematic.

3.2 Late Cretaceous Upper Series

Of the two localities sampled, one (L106) yielded mostly erratic directions at high temperatures, so we did not calculate a mean direction for this locality. The other (L107) yielded more easily interpretable directions (Fig. 6). Most samples from this locality contain one component (northwest and down) after heating to 300°C. The mean tilt-corrected direction is similar to the Early Cretaceous direction (Table 1). Because no fold test is available for the Late Cretaceous redbeds and the number of samples is few, we regard the result from the Upper Series as preliminary, not sufficient to stand alone.

3.3 Tertiary Liuchou and Nadu formations

During thermal demagnetization of samples from the Liuchou Formation (L113) (Fig. 7a), about 70% of the natural remanent magnetization (NRM) decays by 100–200°C. Above 450°C



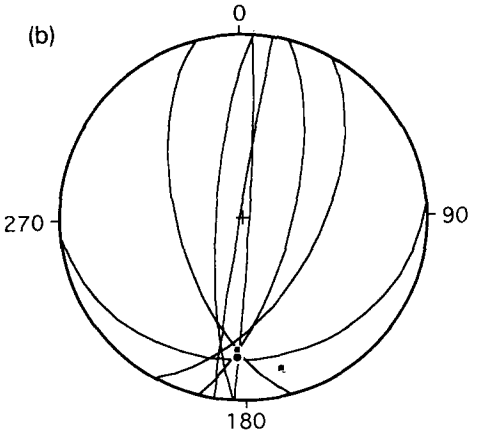


Fig. 7. (a) Thermal vector demagnetization diagram for sample 91U1116A from the Liuchou Formation (L113) (*in-situ* coordinates). ● = vertical component; + = horizontal component; temperature steps are in degrees Celsius. (b) An equal-area plot of demagnetization great circles in *in-situ* coordinates (see text) from seven samples from the Liuchou Formation; ● = mean direction of the nine samples used from the locality; ■ = the two set points.

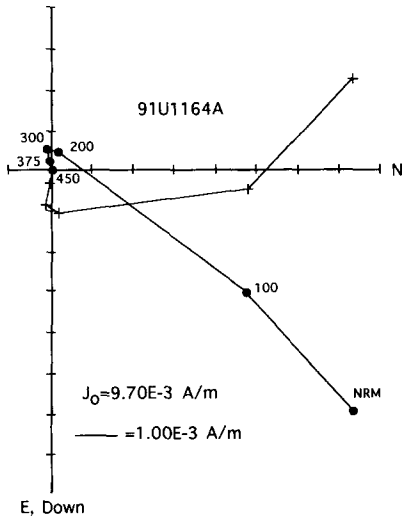


Fig. 8. Thermal vector demagnetization diagram for sample 91U1164A from the Nadu Formation (L119) (*in-situ* coordinates). ● = vertical component; + = horizontal component; temperature steps are in degrees Celsius.

the HTC typically persists in one general direction (e.g., southeast and up) without decaying to the origin. Because of this, we used great circles [26] to help define the mean direction. We generally used the second (100°C) through to the sixth (375°C) demagnetization steps to define the great circles; best-fit line data from two samples were used as set points. The great circles cross at high angles (Fig. 7b) and isolate a mean direction, southeast and up in tilt-corrected coordinates (Table 1). Like the redbed samples from locality L113, all eight siltstone samples from the Nadu Formation (locality L119) have reversed directions after removing an overprint (Fig. 8). The remanence decays to nearly 0% of the NRM by 375–450°C, defining a HTC within the 300–450°C demagnetization steps where the direction decays univectorially to the origin (Table 1).

The analysis of the seventeen samples from the two localities (Table 1) shows that the precision parameter (k) in tilt-corrected coordinates (31.5) is 5.9 times greater than in *in-situ* coordinates (5.3), an increase in clustering that is significant at greater than 95% confidence [24]. The positive fold test and reversed polarities suggest that the mean tilt-corrected direction is a characteristic remanent magnetization from the middle to late Eocene in the Youjiang basin.

3.4 *Late Cretaceous Shaxian Formation*

We thermally demagnetized twenty samples from two localities (L151 and L159) of the Late Cretaceous Shaxian Formation in Fujian Province (Fig. 3). Thermal demagnetization isolates two components: an LTC between 100 and 375°C that trends north and down in *in-situ* coordinates, and an HTC between 375 and 685°C that trends northeast and down in tilt-corrected coordinates and decays univectorially to the origin (Fig. 9). Once again, the high unblocking temperature, the inability of an alternating field at 100 mT to remove an appreciable percentage of the NRM

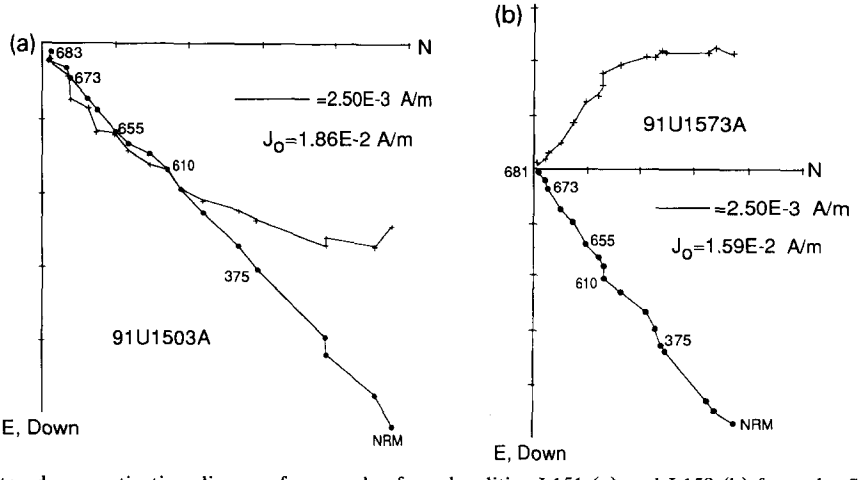


Fig. 9. Thermal vector demagnetization diagram for samples from localities L151 (a) and L159 (b) from the Shaxian Formation (*in-situ* coordinates). ● = vertical component; + = horizontal component; temperature steps are in degrees Celsius.

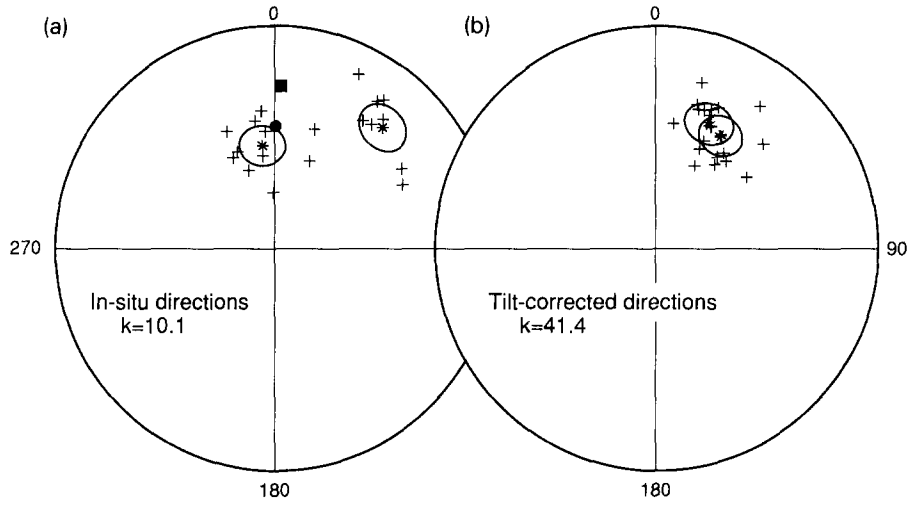


Fig. 10. Equal-area plots of the mean directions and their 95% circle of confidence from the Shaxian Formation before (a) and after (b) unfolding. ■ in (a) represents the present field direction; ● = geocentric axial dipole direction for the present latitude.

and the red color of the samples all suggest that hematite is the magnetic mineral that carries the remanence.

The samples collected from the Shaxian Formation provide a good opportunity to apply the fold test because they come from opposite limbs of a fold. The *in-situ* direction (north and down) and the decrease in k (4.9 times) going from *in-situ* coordinates to tilt-corrected coordinates both imply that the LTC is a recent overprint. In contrast, the precision parameter for the HTC is 4.1 times greater in tilt-corrected coordinates than in *in-situ* coordinates (Table 1 and Fig. 9). The increase in clustering calculated from the samples is significant at the 95% confidence level [24], implying a positive fold test. This conclusion is confirmed by the more rigorous statistical test of McFadden and Jones [27]. Although the directions are all normal, the presence of two components and the stable behavior of the HTC lead us to conclude that the HTC is a near-primary chemical remanent magnetization (CRM) probably acquired during the Cretaceous Long Normal Superchron in the early Late Cretaceous.

The tilt-corrected direction for the Shaxian Formation (Table 1) is similar to the tilt-corrected direction for the Hekou Formation (D = 23.5°, I = 33.5°) reported by Zhai et al. [28]. The samples from both the Shaxian Formation and the underlying Hekou Formation come from the same basin in Fujian Province. We note that although the Hekou Formation was previously reported as Early Cretaceous [29], it is now thought to be Late Cretaceous [20].

3.5 Late Cretaceous Sanshui Formation

We thermally demagnetized nineteen samples from the Sanshui Formation at one locality (L98) (Fig. 11) in Guangdong Province (Fig. 3). As in previous cases, the behavior of these samples in thermal and af demagnetization indicate that the predominant carrier of stable remanence in these red sediments is hematite. The tilt-corrected directions above approximately 500°C are consistently northeast and down, and the *in-situ* incli-

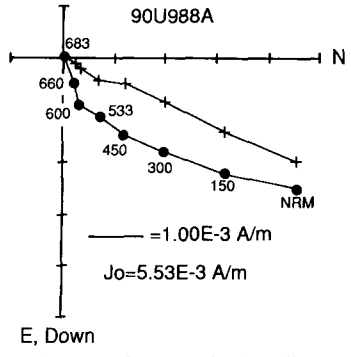


Fig. 11. Thermal vector demagnetization diagram for sample 90U09X8A from the Sanshui Formation (L98) (*in-situ* coordinates). ● = vertical component; + = horizontal component; temperature steps are in degrees Celsius.

TABLE 2: Cretaceous paleomagnetic poles

Name/Area of study	Location on Fig. 1	Lat	Long	Age	λ	Φ	A_{95}	N	Reference
Guangxi Province:									
Xinlong Fm	1	22.7	108.7	K_1	86.5	26.4	6.8	8L	This study
Upper Series	1	22.2	108.7	K_2	79.4	7.1	10.0	9S	This study
Youjiang Basin	2	23.5	107.2	E_1	58.6	18.2	7.4	17S	This study
Nanning Basin	1	22.8	108.4	E_1	83.6	228.0	8.5	5L	Zhao et al. [32]
Sichuan Province:									
Sichuan Basin	A	30.0	102.9	K_1	74.5	229.0	3.7	23L	Enkin et al. [34]
Xichang	B	27.9	102.3	K_1	77.2	196.8	15.3	11S	Zhu et al. [33]
Xiaoba Fm	C	26.5	102.4	K_2	81.5	220.9	7.1	18L	Huang and Opdyke [35]
Feitianshan Fm	C	26.5	102.4	K_1	69.0	204.6	4.3	7L	Huang and Opdyke [35]
Mean Sichuan	-	28.2	102.5	K	76.9	211.6	6.0	5	
Coastal Provinces:									
Shaxian Fm	3	26.0	117.3	K_2	65.1	207.2	5.0	20S	This study
Sanshui Fm	4	23.1	113.3	K_2	56.2	211.5	3.9	19S	This study
Hekou Fm	3	26.0	117.3	K_2	66.9	221.4	5.4	5L	Zhai et al. [28]
No Guangdong Basin	-	24.3	114.8	K_2	66.0	221.5	3.4	43S	Hsu [31]
Mean Coastal Provinces	-	24.3	114.8	K_2	63.7	215.0	6.7	4	

Lat = degrees north latitude of locality; *Long* = degrees east longitude of locality. K = Cretaceous; E = Tertiary; 1 = early; 2 = late. λ = degrees north latitude of pole; Φ = degrees east longitude of pole; A_{95} = radius in degrees of the circle of 95% confidence about the mean pole; N = number of localities (L) or samples (S) used to determine pole

nations are steeper than the tilt-corrected inclinations (Fig. 10). The strikes and dips vary just enough within this locality to apply the McFadden [30] fold test, a more rigorous fold test than the classic McElhinny [24] version, which yields an inconclusive result in this case. Using definition two of the test statistic, the *in-situ* and tilt-corrected correlation values are 5.801 and 0.444, respectively, demonstrating passing of the fold test at 95% confidence. Because of the positive test, we regard the tilt-corrected direction (Table 1) for L98 as a primary CRM, but emphasize that this result can be improved by further sampling. We note that a study by Hsu [31] from a Cretaceous redbed basin in northern Guangdong Province yields a direction that is similar, though reversed ($D = 203.7°$, $I = -27.1°$, $\alpha_{95} = 3.9°$, $N = 43$), to the one reported here for the Sanshui Formation.

4. Discussion

4.1 Paleomagnetic poles

The paleomagnetic poles, calculated from the average of the locality mean directions in the case of the Xinlong Formation and the average of the individual sample directions from the other formations described above, are listed in Table 2. This table also lists another Eocene pole from Guangxi Province [32], four Cretaceous poles from Sichuan Province [33–35], and two Late Cretaceous poles from Guangdong and Fujian provinces [28,31]. The result from Enkin et al.



Fig. 12. Equal-area plot of south China paleomagnetic poles (data from Table 2). Unfilled A_{95} circles are Cretaceous poles from this study. Tertiary $A95$ circles are lightly shaded. The averaged Cretaceous pole from Sichuan and the averaged Cretaceous coastal provinces pole are densely shaded. Note that the Youjiang basin is rotated $40° \pm 8°$ counterclockwise with respect to the Nanning basin, the Xinlong Formation is rotated $18° \pm 8°$ counterclockwise with respect to Sichuan Province, and that the coastal provinces are rotated $12° \pm 8°$ clockwise with respect to Sichuan Province.

[34] (including an 8° rotation) supersedes studies by Enkin et al. [36], Otofuji et al. [37] and Zhuang et al. [38] [see 34]. The Xiaoba Formation pole from Huang and Opdyke [35] in Table 2 supersedes the pole given by Kent et al. [39] [see 35]. The equations and methods used to determine the rotation and latitudinal displacement (and their errors) are found in Coe et al. [40].

The poles from the various localities in this study, the mean pole from both Sichuan Province and the coastal provinces (Fujian and Guangdong), and the Tertiary pole from the Nanning basin of Guangxi Province [32], are graphically displayed in Fig. 12. Several striking features are apparent. The Early Tertiary pole from the Youjiang basin (this study) is rotated counterclockwise ($40° \pm 8°$), but not significantly displaced ($-0.6° \pm 8°$), with respect to the Early Tertiary pole of Zhao et al. [32]. The Cretaceous poles are also streaked out, with little or no significant latitudinal displacement relative to one another. The Xinlong (Guangxi) and Sanshui (Guangdong) poles are end members, with the other Cretaceous poles lying within the streaked zone.

The streaking of the Cretaceous poles apparent in Fig. 12 discourages us from taking the average of currently available poles as the Cretaceous reference pole for south China. The data are too inhomogeneously distributed and the streaking too great to ensure that such an average is representative. We prefer instead to use the average of the poles from Sichuan Province (excluding poles superseded by later studies) as a reference for comparing our results (see Table 2).

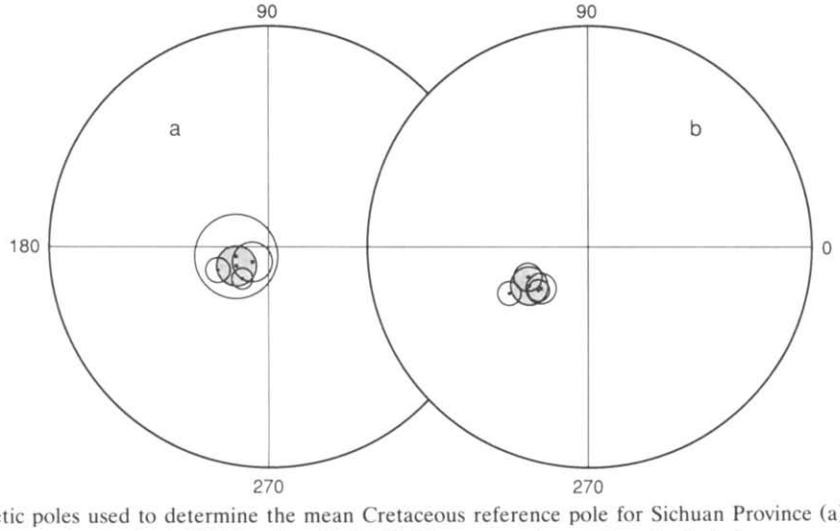


Fig. 13. Paleomagnetic poles used to determine the mean Cretaceous reference pole for Sichuan Province (a) and for the coastal provinces (b) (data from Table 2). The mean poles are densely shaded.

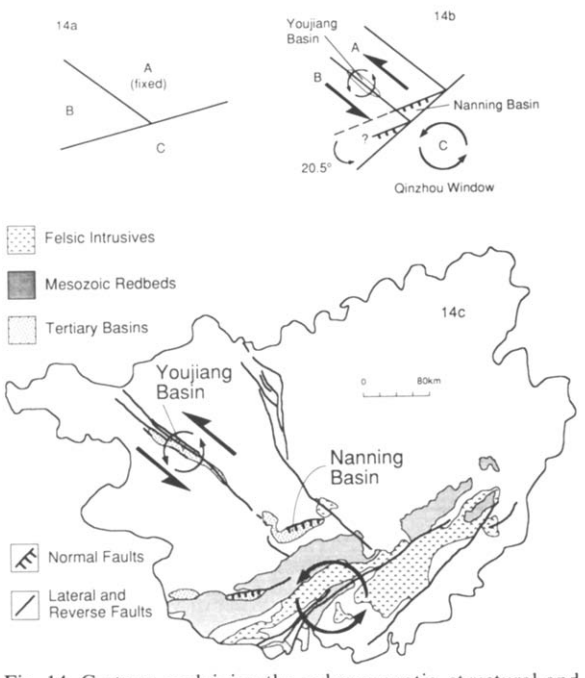


Fig. 14. Cartoon explaining the paleomagnetic, structural and geologic data from Guangxi Province (structural and geologic data after GXBGMR [19]). See text for discussion.

More paleomagnetic studies have been done in Sichuan than in any other province of south China, the poles are not badly streaked (Fig. 13a), and Sichuan is indisputably part of the ancient core (Yangtze craton) of south China. With respect to this mean Sichuan Cretaceous reference pole, the pole for the Xinlong Formation in Guangxi Province, for example, shows $18° \pm 8°$ counterclockwise rotation. The Shaxian Formation in Fujian Province shows $12° \pm 7°$ clockwise rotation relative to the same pole.

4.2 Cenozoic rotations

Enkin et al. [13] drew attention to the streaking of Chinese Cretaceous poles about small circles centered in China. They proposed that this was likely to be the result of local rotation of blocks about vertical axes—i.e., not the result of tectonic accretion between blocks. In support of this notion, the geologic map of Guangxi Province suggests a simple block rotation model (Fig. 14) that fits our paleomagnetic results very well.

As illustrated in Fig. 14, displacement of block B southeast relative to block A along a left-lateral, strike-slip fault (the Youjiang fault) could cause smaller blocks in the Youjiang basin, which lie within the fault zone, to rotate counterclockwise. That such a rotation occurred is demonstrated by our paleomagnetic results from Early Tertiary sediments in this basin, which are rotated $40° \pm 8°$ counterclockwise. The movement of block B could also force block C to rotate counterclockwise about a pivot point on its boundary at the southeast end of block A. This supposition is confirmed by our paleomagnetic result from the Early Cretaceous Xinlong Formation indicating $18° \pm 8°$ counterclockwise rotation. The latter rotation would open extensional basins, such as the Nanning basin, along the northwestern boundary of block C, southwest of the pivot point. No rotation is expected for sediments deposited in these basins, and little was observed by Zhao et al. [32] (see Table 2 and Fig. 12 in this paper) for the Early Tertiary sediments of the Nanning basin, either with respect to the mean Cretaceous Sichuan pole or with respect to a mean Tertiary pole for all China.

Because we do not see evidence of comparable post-Cretaceous compression east of the pivot point, we would not expect significant counterclockwise rotation there. One paleomagnetic sampling locality (L138 in Fig. 2) does lie east of the pivot point by about 100 km, and it does not show any rotation (Table 1). Moreover, we would expect the Late Cretaceous Upper Series paleomagnetic sampling locality (L107) to be rotated counterclockwise by the Tertiary block rotation model that we propose, and indeed it is, by 24°. Eliminating locality L138 and adding L107 to the Xinlong Formation localities, we arrive at a best estimate for post-Cretaceous tectonic rotation of block C of $20.5° \pm 6.5°$.

The block rotation model also correctly predicts the left-lateral displacement of 20 km on the Youjiang fault inferred from the most recent geological map (1:1,000,000 [19]). This measure of fault displacement is derived from the width of the Nanning basin, which we interpret to have formed to accommodate motion on the fault. Using the distance of 65 km from the pivot point to the Youjiang fault and the paleomagnetically inferred rotation of block C, we would predict the fault displacement to be 24 ± 7 km.

The sinistral sense and timing of shear on the Youjiang fault is consistent with lateral escape models for the Himalayan collision [e.g., 41,42]. These call for extrusion of Indochina along the Red River fault system by major, left-lateral strike-slip fault displacement. The Youjiang fault roughly parallels the Red River fault a few hundred kilometers to the northeast (Fig. 2), and the relatively minor Cenozoic motion along it might also be a consequence of the Himalayan collision. If so, more detailed sampling of the Tertiary deposits in the Youjiang basin could help constrain the timing of when the effects of the collision of India with Asia were felt in south China.

Cretaceous sediments sampled in the coastal provinces of Fujian and Guangdong are rotated less and in the opposite sense (Fig. 12). The clockwise rotation is consistent in direction and magnitude with other results from the area [28,31] (Fig. 13b). This may indicate that the region from Guangdong to Yong'an rotated as a single block, $12° \pm 8°$ clockwise with respect to Sichuan Province (Table 2 and Fig. 12). This would be consistent with the predicted lateral escape of south China resulting from the India–Asia collision [41]. However, the number of localities in the region is small and most of the studies are preliminary. The possibility remains that the results reflect local block rotations, but in this case driven by circum-Pacific tectonics. Focal plane solutions of recent earthquakes in the north China basins appear to represent pull-apart motion associated with right-lateral displacement on north-northeast-trending faults [43]. Focal plane solutions from small earthquakes in Guangdong Province in 1981 also show right-lateral motion occurring along northeast-trending strike-slip faults [44]. Similar faults are mapped in Fujian Province (Fig. 3), and although seismically quiescent today were probably formed by the same stress regime as that acting in Guangdong. A systematic paleomagnetic study, sampling a wide distribution of the basins in the region, should be able to distinguish between the two possibilities.

5. Conclusions

The streaking of Cretaceous paleomagnetic poles obtained from samples in southeast China is caused by the local rotation of blocks, as concluded earlier by Enkin et al. [13] for the whole of China. We find that the Xinlong and Upper Series formations of southern Guangxi Province are rotated counterclockwise, whereas the Shaxian Formation of central Fujian Province and the Sanshui Formation of southeastern Guangdong Province are rotated clockwise with respect to a mean Cretaceous reference pole for Sichuan Province. In Guangxi Province, Cenozoic left-lateral displacement on a southeast-trending fault, probably related to the Himalayan orogeny, has rotated a small basin in the fault zone, and may at the same time be responsible for the rotation of the much larger, northeast-trending block against which it abuts. In Fujian and Guangdong provinces, similar magnitudes of clockwise-rotated Cretaceous basins might reflect a regional block rotation compatible with the extrusion model. However, the observed pattern could also result from right-lateral shear acting parallel to the continental margin and driven by circum-Pacific tectonics.

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